

Problem-Solution Fit Canvas

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Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS]	2. PROBLEMS / PAINS + FREQUENCY [PR]	3. TRIGGERS TO ACT [TR]
• Smallholder and commercial farmers (primary) • Agricultural extension officers and agronomists • Agricultural researchers and policy makers • Agri-tech startups and NGOs serving rural India	• Wrong crop selection causes financial loss (every season) • Plant disease undetected until crop is visibly damaged (frequent) • Over/under-fertilization wastes money and degrades soil (every cycle) • No instant, free weather data for soil input forms (constant pain)	• Crop failure or low yield in the previous season • Rising fertilizer costs pushing farmers to optimise usage • Government push for data-driven precision agriculture • Awareness of free AI tools via smartphones
6. CUSTOMER LIMITATIONS [CL]	9. PROBLEM ROOT CAUSE [RC]	7. BEHAVIOR + INTENSITY [BE]
• Limited smartphone or computer literacy in rural areas • Low internet bandwidth in villages • Soil testing cost ₹500–₹2000 per test per season • No access to agronomists in remote districts	• No unified platform integrating soil, climate, and crop recommendation data • Agricultural extension services are under-staffed and under-resourced • Existing ML tools require cloud accounts, API keys, or app installs	• Rely on neighbours or village elders for crop advice (High intensity, every season) • Use government booklets for disease identification (Medium, occasional) • Visit an agri officer only when crop is already failing (Low frequency)
5. AVAILABLE SOLUTIONS — PROS & CONS [AS]	10. YOUR SOLUTION [SL]	8. CHANNELS OF BEHAVIOR [CH]
Pros: Some govt crop advisory apps exist; Krishi Vigyan Kendras provide advice. Cons: Not integrated; require separate logins; no real-time weather, no disease image detection, no fertilizer guidance, no analytics — all in one offline-capable tool.	OptiCrop: A single, no-login, local Python web app with: • ML crop recommendation (RF+DT+SVC ensemble) • Symptom + image plant disease detection • Crop suitability assessment with remediation steps • Fertilizer advisor with NPK gap analysis • Open-Meteo real-time weather auto-fill • Research analytics dashboard + CSV export	ONLINE: • Web browser (Chrome, Firefox, Edge) • GPS weather auto-fill via browser geolocation API • Open-Meteo API for live weather data OFFLINE: • Fallback JSON dataset when CSV unavailable • Graceful error responses when API is unreachable
4. EMOTIONS BEFORE / AFTER [EM]		

BEFORE using OptiCrop: Anxious, uncertain, financially stressed, reliant on guesswork and outdated advice. AFTER using OptiCrop: Confident, data-informed, empowered to select the right crop, apply the right fertilizer, and detect disease early — all from a free, no-login tool on their local machine.		
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